

PIYUSH PAUL

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PROFESSIONAL SUMMARY

Computer Science undergraduate (AI/ML specialization) with hands-on experience building AI-powered SaaS platforms, autonomous agent systems, and high-performance backend services. Proficient in Python, FastAPI, Flask, LLM integration, multi-agent orchestration, and cloud deployment on GCP. Proven ability to ship production-quality systems under hackathon conditions. Seeking internship and early-career roles in AI/ML Engineering or Backend Engineering.

TECHNICAL SKILLS

Programming Languages: Python, JavaScript, TypeScript, C, C++, Java, SQL

AI / ML: LLM integration (Gemini, OpenAI GPT, Anthropic Claude), multi-agent orchestration, RAG, prompt engineering, NLP, agentic workflow design

Frameworks & APIs: FastAPI, Flask, Next.js, REST API design, SSE architecture, microservices

Cloud & DevOps: GCP, Docker, Git, CI/CD

Databases: MongoDB, PostgreSQL, SQLite, Supabase

Other: DPDK, multi-threaded systems, network programming

PROFESSIONAL EXPERIENCE

Backend Developer Intern — Fluencer Market (Remote)

Dec 2024 – Apr 2025

- Designed and developed Flask REST APIs to support influencer-brand collaboration workflows used in production
- Reduced backend API latency by 35% through optimized database queries and response caching strategies
- Collaborated with cross-functional engineering teams to deliver scalable backend services on schedule

PROJECTS

ARTA — Autonomous Red Team Agent (Multi-Agent AI)

Apr 2026

Technologies: FastAPI, Gemini AI, SQLite, Puppeteer, Nmap, Metasploit | GitHub: github.com/Piyush800x/Arta

- Built an AI-driven autonomous penetration-testing pipeline using Gemini; multi-agent system coordinating target discovery, CVE analysis, and exploit validation sub-agents in hierarchical workflows
- Implemented Server-Sent Events (SSE) for real-time asynchronous log streaming and live findings visualization
- Integrated LLM orchestration with Nmap (scanning) and Metasploit (exploitation), maintaining tight context control across agent handoffs
- Won MLH Hack Days 2026, validating ability to ship production-quality AI agent systems in time-constrained environments

NeighbourNet — Offline Mesh Networking App (Disaster Communication)

Mar 2026

Technologies: React Native, BLE/WiFi Direct, mesh protocol

- Built an offline-first mesh networking application enabling disaster communication without internet infrastructure
- Won 1st Runner-Up (Gold) at HackTropica 2026 (MLH-sanctioned hackathon) competing against 20+ teams

AstroRoni — Full-Stack AI SaaS Platform (Cloud / ML Integration)

Sep 2025 – Apr 2026

Technologies: Next.js, Flask, MongoDB, Razorpay, OpenAI, GCP | Link: astroroni.com

- Designed and deployed microservices architecture on GCP handling authentication, payment gating, and AI-driven report generation via OpenAI APIs
- Built RESTful APIs and data models supporting concurrent multi-user LLM-powered workflows
- Integrated LLM inference into a production pipeline with async result delivery and Razorpay subscription billing

PacketForge — High-Performance Network Systems Pipeline

May 2026

Technologies: C, DPDK, Multi-threading, FastAPI, Next.js, QEMU/KVM

- Architected a dual-NIC multi-threaded packet-processing pipeline sustaining 470,000 packets per second (pps)
- Built a real-time telemetry dashboard (FastAPI + React) for live performance monitoring
- Won 1st Runner-Up at Synchronicity S2 (24-hour hackathon, Jadavpur University ACM Chapter)

ACHIEVEMENTS & AWARDS

- MLH HackTropica 2026 — 1st Runner-Up (Gold): Built NeighbourNet, an offline mesh-networking solution for disaster communication
- MLH Hack Days 2026 — Winner: Built ARTA, an AI-powered autonomous multi-agent penetration-testing system
- Synchronicity S2, Jadavpur University — 1st Runner-Up: Built PacketForge, a DPDK-based high-performance DDoS scrubber sustaining 470K pps

EDUCATION

JIS College of Engineering, West Bengal, India

Jun 2025 – Present (Expected 2028)

Bachelor of Technology in Computer Science (AI/ML Specialization) | SGPA: 7.9 / 10

Relevant Coursework: Data Structures & Algorithms, Operating Systems, Database Management Systems, Machine Learning

ADDITIONAL

- Open to relocation and remote internship opportunities across India
- Active hackathon competitor: 3 MLH wins including HackTropica 2026, MLH Hack Days 2026
- Languages: English (professional), Bengali (native), Hindi (conversational)